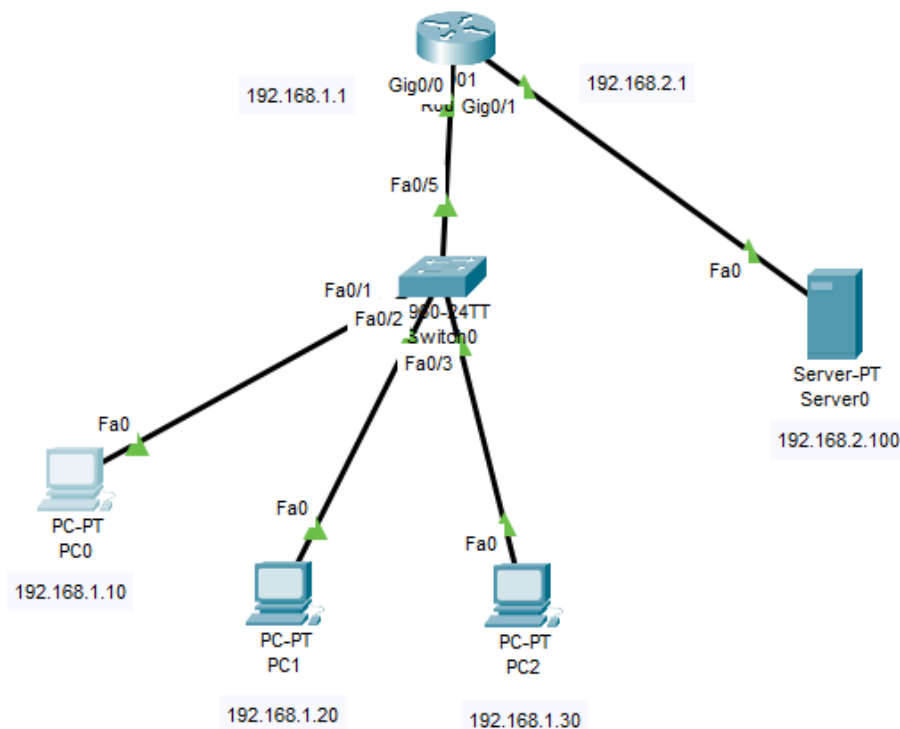


Procedure of CNS Experiment 3

Aim: To demonstrate how to configure **Standard and Extended ACLs** on a router to control which remote hosts can access a server. The goal is to allow some IP addresses while blocking others.

Step-by-Step Implementation

Step 1: Create Network Topology



Purpose: To simulate a LAN where multiple hosts attempt to access a server through a router.

Step 2: Assign IP Addresses

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.20	255.255.255.0	192.168.1.1
PC3	192.168.1.30	255.255.255.0	192.168.1.1
Server	192.168.2.100	255.255.255.0	192.168.2.1
Gig0/0	192.168.1.1	255.255.255.0	
Gig0/1	192.168.2.1	255.255.255.0	

Step 3: Configure Router Interface

Open Router CLI:

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

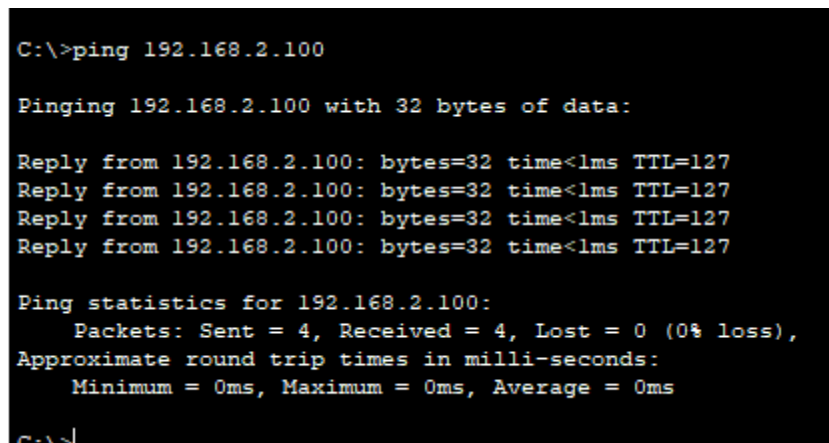
Purpose: Enables routing between hosts and acts as default gateway.

Step 4: Verify Connectivity

From PC0:

```
ping 192.168.2.100
```

Expected: Successful ping replies. It ensures the network works before applying security restrictions.



```
C:\>ping 192.168.2.100

Pinging 192.168.2.100 with 32 bytes of data:

Reply from 192.168.2.100: bytes=32 time<1ms TTL=127
Reply from 192.168.2.100: bytes=32 time<1ms TTL=127
Reply from 192.168.2.100: bytes=32 time<1ms TTL=127
Reply from 192.168.2.100: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PART A — Standard ACL Configuration

Step 5: Create Standard ACL (Block Specific Host)

Standard ACL filters based only on **source IP address**.

Example: Block PC2 (192.168.1.20) from accessing the server

```
access-list 10 deny 192.168.1.20 0.0.0.0
access-list 10 permit any
```

- deny 192.168.1.20 → Blocks that host
- permit any → Allows all others
- 0.0.0.0 → Wildcard mask: {Here 0.0.0.0 Means: Match this exact IP only}
 - A **wildcard mask** is a special mask used in Access Control Lists (ACLs) to tell the router which part of an IP address to check and which part to ignore when filtering traffic.

- Bit value 0 means “Must match exactly”, and bit value 1 means “Ignore (Can be anything)”
- Here 192.168.1.20 0.0.0.0 Means: Match this exact IP only
- In access-list 10 deny 192.168.1.0 0.0.0.255 Meaning is: Match any host from 192.168.1.0–192.168.1.255

Step 6: Apply ACL to Router Interface

- Apply the ACL on client-side interface,
- Direction: IN (traffic entering router from PCs)

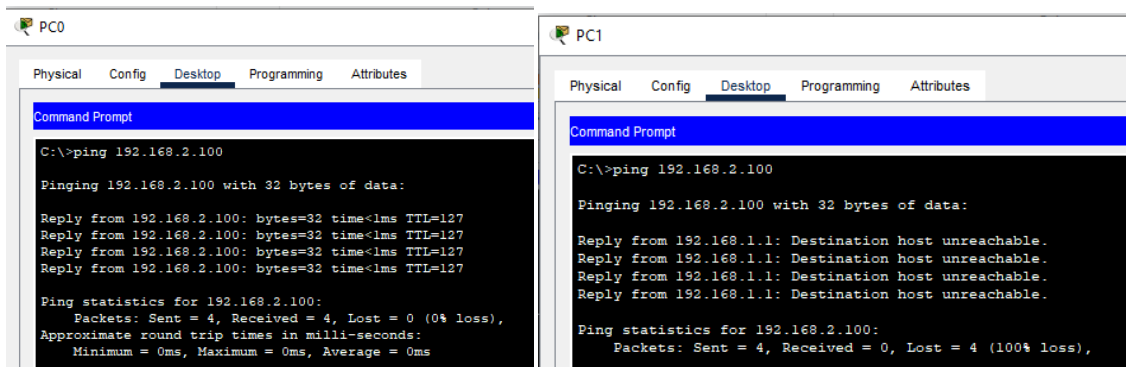

```
interface g0/0
ip access-group 10 in
exit
```

Purpose: Activates filtering on incoming traffic.

Step 7: Test Standard ACL

From PCs:

- PC1 → Ping Server → Should succeed
- PC2 → Ping Server → Should fail



PART B — Extended ACL Configuration

Extended ACL filters by: Source IP, Destination IP, Protocol, and Port numbers (HTTP, FTP, etc.)

Step 8: Create Extended ACL

Example: Allow only HTTP access from PC1 to Server, Deny FTP access from PC2

```
access-list 100 permit tcp host 192.168.1.10 host 192.168.2.100 eq 80
access-list 100 deny tcp host 192.168.1.20 host 192.168.2.100 eq 21
access-list 100 permit ip any any
```

- The first allows only PC1 (192.168.1.10) to access the server (192.168.2.100) using the HTTP service, which operates on TCP port 80.
- The next line blocks PC2 (192.168.1.20) from accessing the server via FTP, which uses TCP port 21.

- Finally, the statement access-list 100 permit ip any any allows all other types of IP traffic not explicitly denied by previous rules.

Command Part	Meaning
tcp	Protocol
host	Single IP
eq 80	HTTP port
eq 21	FTP port

Step 9: Apply Extended ACL

Apply close to the source:

```
interface g0/0
no ip access-group 10 in
ip access-group 100 in
exit
```

- The Extended Access Control List is applied to the appropriate router interface so that it can actively filter network traffic.
- Before applying the new Extended ACL, the previously configured Standard ACL is removed from the interface, because only one ACL can be applied per interface per direction at a time.
- Then the Extended ACL (number 100) is activated on interface GigabitEthernet0/0.

Step 10: Configure Server Services

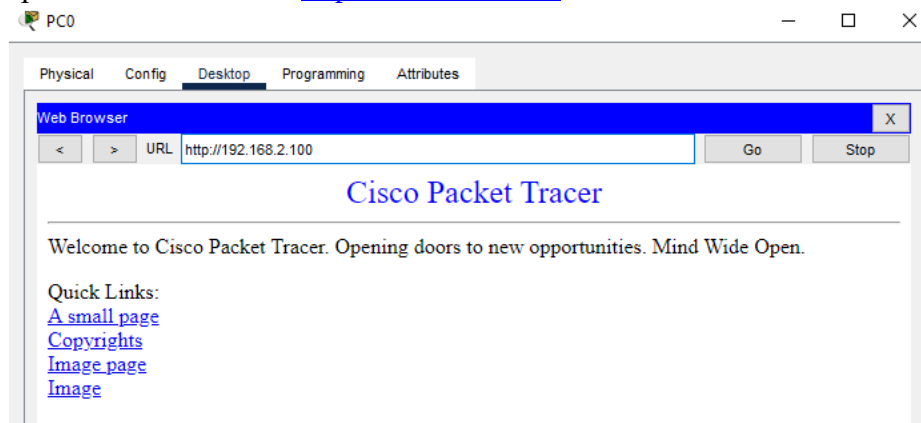
On Server:

- Desktop → IP Configuration (already set)
- Services Tab:
 - Enable HTTP
 - Enable FTP

Step 11: Test Extended ACL

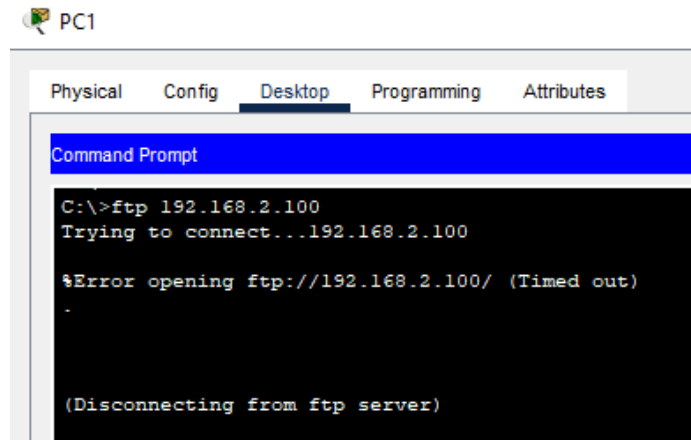
From PC0

- Open Web Browser → <http://192.168.2.100> → Allowed



From PC1

- Try FTP → Denied



Step 12: Verify ACL Configuration

On Router CLI:

```
show access-lists
show ip interface
show running-config
```

Purpose: Confirms rules are applied and working.